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A study on Raman spectroscopic scheme enabling fast and accurate determination of calcite concentration in gypsum

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ABSTRACT

This study intended to develop a potential Raman spectroscopic measurement scheme able to simply and accurately determine the calcite concentration (impurity) in gypsum. The sample coverage area for measurements was a major concern since it directly influenced on the representation of sample composition. Initially, 16 synthetic gypsum samples containing varying calcite contents (0.01 ~ 5.1 wt%) were prepared. For Raman spectral collection, both small spot illumination (50 μm in diameter) and wide area illumination (WAI, 6 mm in diameter) schemes were employed. When the microscope-based (small spot) scheme was used for the analysis, the accuracy of calcite concentration determination was not satisfactory since the sample coverage area was too small to average out the local compositional heterogeneity. The accuracy substantially improved by using the WAI-Raman measurement due to better representation of calcite concentrations in the samples. Next, when real process gypsum samples containing other minor impurities such as carbon residues were analyzed, the fluorescence background appeared due to the colored impurities. Nonetheless, the calcite peak clearly presented on top of the fluorescence background and thereby determination of calcite concentration was feasible. Again, the accuracy of WAI-Raman measurement was superior to the small-spot measurement. In overall, the more correct representation of calcite concentrations in gypsum samples by the WAI scheme was responsible for the improved accuracy.

1. Introduction

Gypsum ($CaSO_4{\cdot}2H_2O$) is a popular material widely employed as a raw material in diverse forms of plaster, wallboard, and chalk [1–5]. One of the main industrial processes producing gypsum is wet flue-gas desulfurization (FGD, $SO_2 + CaCO_3 \rightarrow CaSO_4$), which removes SO_2 in exhaust flue gases from fossil-fuel power plants by scrubbing with an alkali solid slurry composed of $CaCO_3$ (calcite, limestone) [6–9]. As the SO_2 removal efficiency of wet FGD is around 95% or higher, a small amount of calcite exists in the gypsum product. Therefore, determination of calcite content in the final gypsum is necessary for monitoring FGD efficiency in a power plant. Also, since impurities including calcite in the gypsum clearly affect the final quality of subsequent gypsum-containing products, analysis of calcite concentration is essential for quality control of related products.

The most common analytical method for determination of calcite content in gypsum is thermogravimetric analysis (TGA) [10–13], based on the weight measurement of generated CO_2 from thermal decomposition of calcite ($CaCO_3 \rightarrow CaO + CO_2$) at higher than 600 °C. However, TGA requires a long analysis time, typically longer than 50 min, due to the continual change of sample temperature. Therefore, an alternative analytical method enabling simple, fast, and non-destructive determination of calcite content in gypsum is in great demand. In this context, Raman spectroscopy could be a potential candidate since it provides distinguishable molecular vibrational information of especially inorganic compounds and its measurement is rapid.

With this momentum, two separate Raman measurement schemes with different laser illumination areas of small spot illumination (50 μm in diameter) and wide area illumination (WAI, 6 mm in diameter) [14] were comparatively employed for the analysis. As known, a sample-coverage (laser illumination) area in Raman measurement is an important factor governing representation of sample composition, which is ultimately related to accuracy of subsequent quantitative analysis. Therefore, the two different schemes would result in dissimilar accuracies in the determination of low-level calcite content in gypsum. For initial evaluation, 16 synthetic gypsum pellets containing calcites in a

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range from 0.05 to 5.1 wt% were prepared, and back-scattering Raman spectra were acquired by using the small spot illumination as well as WAI schemes. In each measurement, band shapes of calcite, gypsum, and calcite concentration-dependent spectral variation were examined. Then, the concentrations of calcite in the samples were determined using a peak area ratio method, and subsequent accuracies in both cases were compared.

Next, real process gypsum samples were measured. The process samples are grey and khaki, while the synthetic samples are white (refer to Figure S1 in the supplementary information). Therefore, the occurrence of fluorescence, which obscures calcite Raman peaks, would be probable in the Raman measurements. The calcite concentrations in the process gypsum samples were determined using each scheme and compared with those measured by TGA. Finally, the overall results of both cases were comparatively discussed.

2. Experimental section

2.1. Preparation of synthetic mixtures and collection of real process gypsum samples

High purity calcite (99.0%) and gypsum (99.0%) were purchased from Sigma-Aldrich. Sixteen binary mixtures of calcite and gypsum (calcite concentration: 0.05 ~ 5.1 wt%) were prepared by mixing desired amounts of the components. Each mixture sample was hand-shaken for 10 min and vortexed for 10 min to ensure homogeneous mixing. From each mixture sample, a 350 mg sample was taken to make the corresponding pellet using a presser (PIKE Technologies) by applying 10 ton/cm^2 for 2 min. A photograph in Figure S1 (a) shows all the synthetic mixture pellets, all of which were white.

Seventeen real process gypsum powder samples were provided by the KCC, Korea. Again, a photograph of the pellets prepared from the process samples is shown in the supplementary information (Fig. S1 (b)). Dissimilar to the synthetic mixture pellets (Fig. S1 (a)), the process samples greatly differ in color such as grey and khaki. To determine calcite content in the process samples using TGA (Perkin Elmer, TGA 7), a 10 mg of sample was introduced into the instrument, and temperature was increased from 20 to 1000 °C at a rate of 20 °C/min during the analysis. As mentioned earlier, calcite content was determined by measuring the weight of CO_2 generated from thermal decomposition of calcite in the sample.

2.2. Acquisition of Raman spectra of gypsum samples

For Raman spectral acquisition, each sample pellet was positioned on a microscope stage connected with a Raman spectrometer (Kaiser Optical Systems, USA), and laser radiation (785 nm) was focused using an objective lens (10 × 0.25 numerical aperture (NA)) to collect corresponding spectra. The laser spot size was approximately 50 μm in diameter (small-spot scheme). For each sample, 10 replicate spectra (resolution: 4 cm^{-1}) were collected by laser illumination at 5 spots on the top and bottom faces of the pellet, and the pellet was continuously rotated (3 rotations per minute) during spectral acquisition to increase sample coverage area. Each Raman spectrum was acquired as an accumulation of 16 scans with 5-second exposure at each (total acquisition time: 80 s). Also, Raman spectra of the same samples were collected using the WAI scheme (PhAT probe, Kaiser Optical Systems) providing a circular laser illumination spot with a diameter of 6 mm to cover a large sample area (28.3 mm^2). For each sample, 6 replicate spectra were collected by laser illumination at 3 locations on the top and bottom faces of the pellet without sample rotation. Baseline correction of all collected spectra based on asymmetric least squares smoothing (AsLS) [15–17] and peak area calculation were performed using MATLAB version 2020a (The Math Works Inc., MA, USA).

3. Results and discussion

3.1. Measurement of calcite concentration in synthetic and process gypsum samples using micro-Raman spectroscopy

Fig. 1 (a) presents the Raman spectra of pure gypsum and calcite in the 1600–400 cm^{-1} range acquired using the Raman microscope with the laser illumination diameter of 50 μm (small-spot scheme). Six peaks from gypsum are observed at 1136, 1008, 672, 618, 497, and 413 cm^{-1}, and the 1008 cm^{-1} peak is most intense [18,19]. In the calcite spectrum, there is a strong peak at 1087 cm^{-1}, and two less intense peaks appear at 1440 and 711 cm^{-1} [18,19]. Since the intense calcite peak at 1087 cm^{-1} does not overlap with the gypsum peaks, determination of calcite content in gypsum is not expected to be complicated, while acquisition of calcite content-representative spectra is a prerequisite to secure accuracy.

Fig. 1 (b) presents the 10 replicate Raman spectra of 1.6%-calcite synthetic mixture (pellet) in the 1170–1070 cm^{-1} range after treatment of AsLS. The raw spectra before treatment of AsLS are presented in the supplementary information (Figure S2). The calcite peak at 1087 cm^{-1} and gypsum peak at 1136 cm^{-1} are present in this range. Ten replicate spectra were collected by laser illumination of the sample at 5 spots on the top as well as bottom faces of the pellet, while the pellet was continuously rotated during spectral acquisition as described earlier. As seen, the intensities of both calcite and gypsum peaks moderately vary. To estimate reproducibility, the areas of calcite peak under the 1100–1075 cm^{-1} range (A_c) and gypsum peak under the 1170–1100 cm^{-1} range (A_g) were obtained from the 10 replicate spectra, and corresponding A_c/A_g values were calculated. RSD of the 10 A_c/A_g values was 8.3%.

Fig. 1 (c) shows the average Raman spectra (AsLS-treated spectra) of 10 replicate spectra collected from each of the 16 synthetic mixtures. With elevation of calcite concentration, the intensities of the calcite peak at 1087 cm^{-1} and gypsum peak at 1136 cm^{-1} generally increase and decrease, respectively. A notable observation is that the positions of both peaks differ in a few spectra, which indicates local compositional heterogeneity in the samples, and the increased coverage area by sample rotation during spectral collection is not sufficient to average out the local heterogeneity. Fig. 1 (d) presents the correlation between calcite concentration and A_c/A_g value. The correlation is good with R^2 of 0.97. The sample rotation helped to represent the calcite concentration in the sample more accurately. Nonetheless, the data are somewhat scattered along the regression line.

Fig. 2 (a) shows the average Raman spectra of 10 replicate spectra collected from each of the 17 real process samples. Fluorescence is present in many spectra, and the magnitude varies by sample. The fluorescence probably originates from impurities such as carbon residues, responsible for the grey-like sample colors observed in Fig. S1 (b) (refer to the supplementary information). The calcite and gypsum peaks appear on the top of the fluorescence background, while their dominances are relatively weak due to the high fluorescence intensity. Therefore, subtraction of fluorescence background is necessary to highlight the sample peaks. For this purpose, the raw spectra were treated again with AsLS as shown in Fig. 2 (b). Notably, the new peaks at 1105 cm^{-1}, not observed in the spectra of synthetic samples (Fig. 1 (c)), appear beside the calcite peak at 1087 cm^{-1}. The 1105 cm^{-1} peak probably originates from other minor components in the process samples. The positions of peaks, especially the gypsum peak at 1136 cm^{-1}, vary to a small degree as also noticed in the synthetic mixture spectra (Fig. 1 (c)). Fig. 2 (c) shows the correlation between calcite concentration and A_c/A_g value for analysis of the process samples. The calcite concentration on the *x*-axis was determined using TGA. The spectral range used to calculate the calcite peak area was narrowed to 1095–1080 cm^{-1} to exclude the 1105 cm^{-1} peak. In comparison with the result in measurement of the synthetic mixture sample (Fig. 1 (d)), the correlation becomes considerably degraded (R^2: 0.70). For analysis

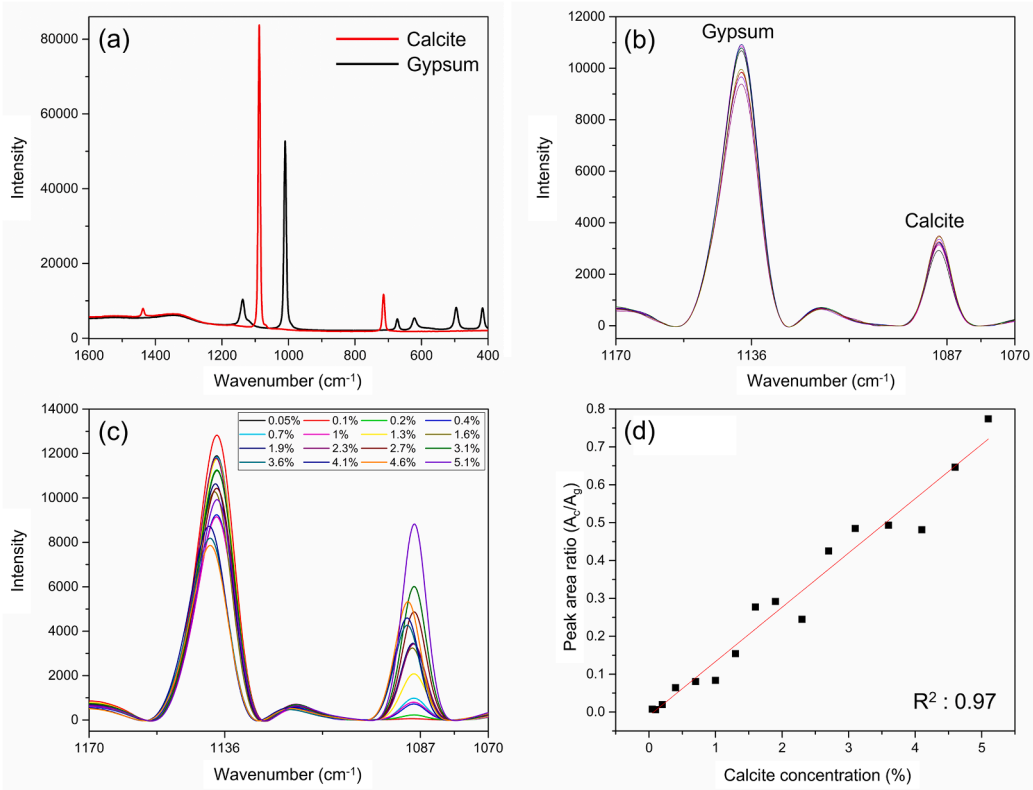

Fig. 1. Raman spectra of pure gypsum and calcite in the 1600–400 cm^{-1} range (a) and 10 replicate spectra of a 1.6%-calcite standard mixture in the 1170–1070 cm^{-1} range after treatment of AsLS (b). Average Raman spectra of all 16 synthetic mixtures collected using the Raman microscope with sample rotation (c) and subsequent correlation between calcite concentration and A_c/A_g value (d) are also shown.

of the process samples, micro-Raman measurement is not reliable enough.

3.2. Measurement of calcite concentration in synthetic and process gypsum samples using WAI Raman spectroscopy

As found in the previous case, a straightforward way to improve accuracy is to increase the sample coverage area in spectroscopic measurement. For this purpose, a wide area illumination (WAI) scheme providing a 6 mm-diameter laser illumination (area: 28.3 mm^2) was adopted. Fig. 3 (a) shows the six replicate Raman spectra (AsLS-treated spectra) of a 1.6%-calcite synthetic mixture collected using the WAI scheme with laser illumination of the sample at three spots on the top and bottom faces of the pellet. The peak intensity variations among the replicate spectra are reduced, especially for the gypsum peak at 1136 cm^{-1}, compared to those in the micro-Raman measurement (Fig. 1 (b)). The RSD of the corresponding 6 A_c/A_g values was 3.4%, indicating improved reproducibility.

Fig. 3 (b) shows the average Raman spectra (AsLS-treated spectra) of 6 replicate spectra collected from each of the 16 synthetic mixtures. The intensities of the 1087 cm^{-1} calcite peak clearly change according to calcite concentration. The positions of both peaks are consistent with the WAI measurement, while they vary in the micro-Raman measurement. Fig. 3 (c) presents the correlation between calcite concentration and A_c/A_g value, which is linear and good (R^2: 0.98). This indicates that the WAI scheme is effective to collect reproducible and representative Raman spectra to determine sample composition.

Fig. 4 (a) and (b) show the average Raman spectra of the 17 process samples and subsequent AsLS-treated spectra, respectively. Again, fluorescence is present in many raw spectra. After AsLS-treatment, the three peaks are apparent, and their positions are consistent. Also, the intensity of the 1087 cm^{-1} calcite peak largely varies due to the change of calcite concentration in the samples. In comparison with the results in the micro-Raman measurement (Fig. 2 (c)), the correlation between calcite concentration and A_c/A_g value vastly improves, with R^2 of 0.90, due to the more correct representation of calcite concentration in the sample by WAI measurement.

Meanwhile, the correlation in the measurement of process samples is inferior to that in the measurement of synthetic mixtures (Fig. 3 (c)), which can be explained in several ways. First, there are minor components other than calcite and gypsum in the process sample, and their spectral features interfere with the neighboring calcite peak. This increases the uncertainty in determination of calcite concentration. Second, only 10 mg of each sample was used for TGA to determine the reference calcite concentration and likely does not represent the overall sample composition, potentially introducing an error. Third, since selective subtraction of only the superimposed fluorescence background in the process sample spectra is difficult, the AsLS-treated spectra might not perfectly convey the real calcite concentration.

4. Conclusions

The WAI Raman scheme providing a 6 mm-diameter laser illumination (coverage area: 28.3 mm^2) was effective to average out the local compositional heterogeneity in gypsum samples and helpful to represent sample calcite concentrations more accurately. Although fluorescence emanated from the colored impurities in the real process samples, the calcite peak clearly appeared on top of the fluorescence background and allowed measurement of calcite concentration. The overall result indicates the WAI Raman scheme as a potential alternative to TGA, the slower conventional analytical method. More sophisticated algorithms for selective subtraction of fluorescence background would be beneficial to improve the accuracy further.

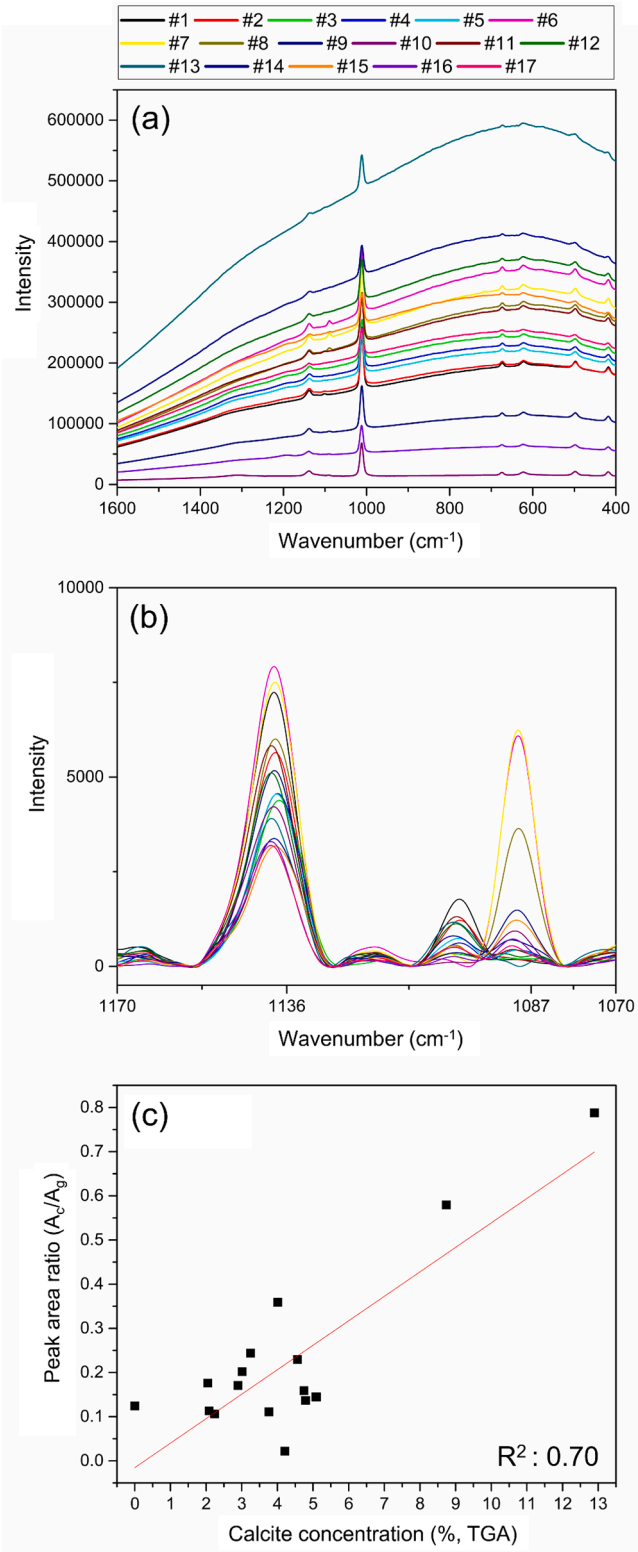

Fig. 2. Average Raman spectra of all 17 process gypsum samples acquired using the Raman microscope with sample rotation (a), subsequent AsLS-treated spectra (b), and correlation between calcite concentration and A_c/A_g value (c).

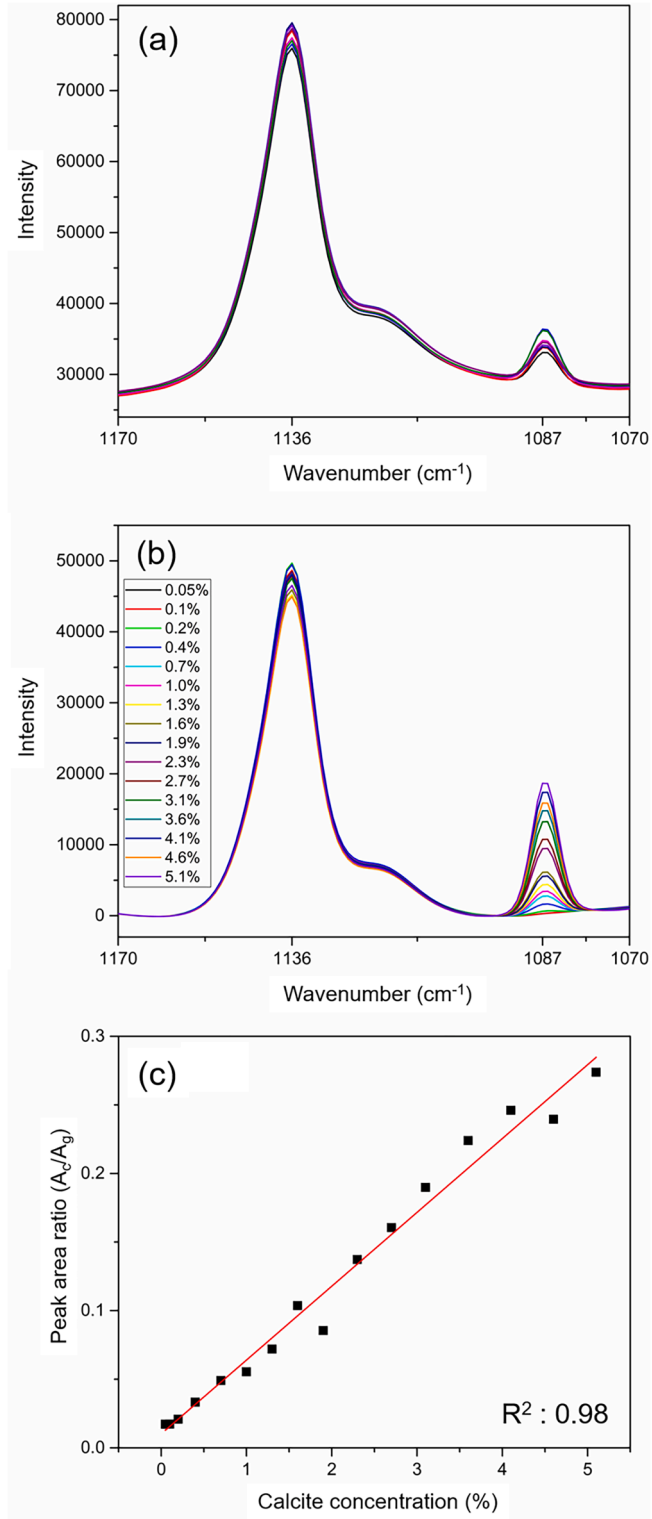

Fig. 3. Raman spectra of a 1.6%-calcite synthetic mixture collected using the WAI Raman scheme (a), average spectra of all 16 synthetic mixtures (b), and subsequent correlation between calcite concentration and A_c/A_g value (c).

CRediT authorship contribution statement

Daun Seol: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – review & editing, Visualization. **Youngtaek Ma:** Conceptualization, Resources. **Kyungmin Nam:** Software, Validation, Formal analysis, Investigation, Data curation. **Hoeil Chung:** Conceptualization, Methodology, Resources, Writing – original draft, Writing – review & editing, Supervision.

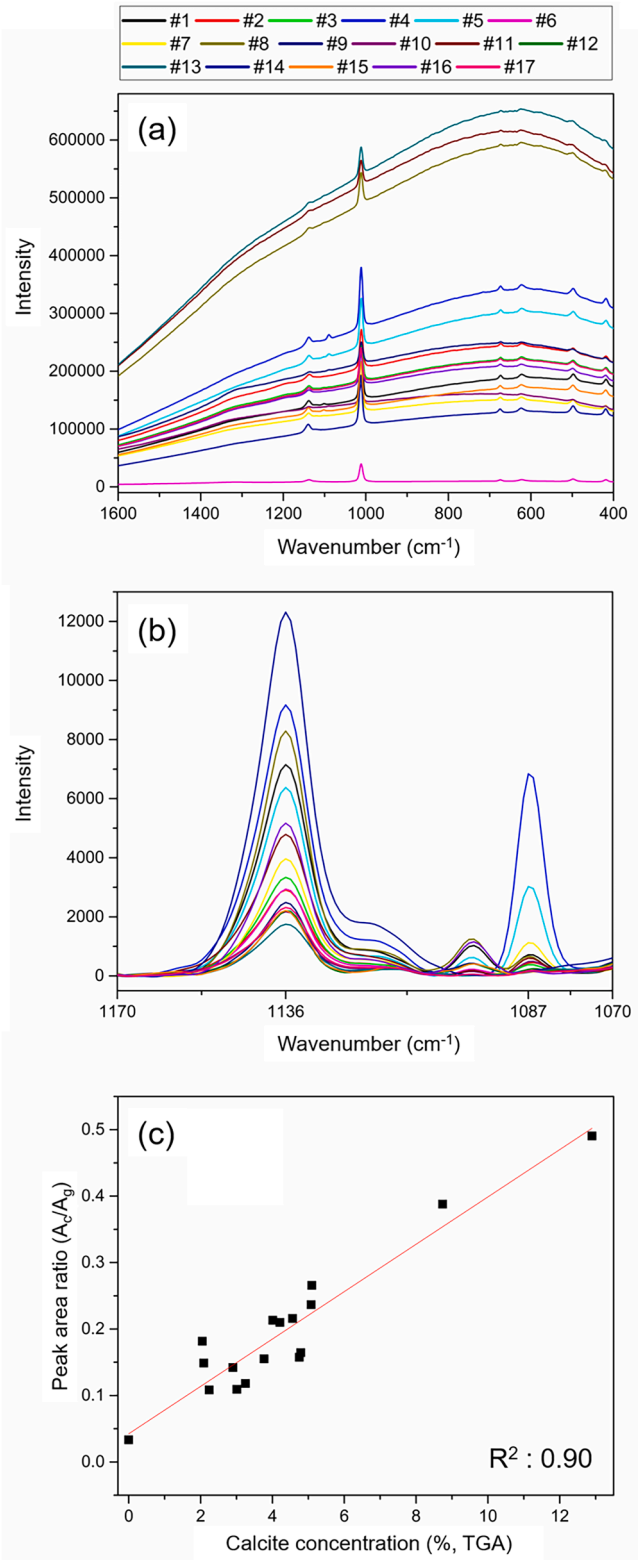

Fig. 4. Average Raman spectra of all 17 process gypsum samples acquired using the WAI Raman scheme (a), subsequent AsLS-treated spectra (b), and correlation between calcite concentration and A_c/A_g value (c).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.microc.2021.106913.

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